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Software Requirements Specification (SRS)

1. Introduction:

The **Hotel Booking System** is designed to help users search for hotels, view detailed hotel information, compare prices from different providers, and book hotel rooms online in a secure and efficient way. The system aims to simplify the hotel selection and booking process by providing accurate and organized information, improving user experience, and integrating with external systems such as booking providers and payment gateways.

This system supports the tourism and hospitality industry by reducing the time and effort required to find and book suitable accommodation

2. User Requirements:

- The user should be able to search for hotels in a specific city.
- The user should be able to apply filters to search results (price, rating, number of stars).
- The user should be able to view detailed information about a selected hotel.
- The user should be able to compare hotel prices from different providers.
- The user should be able to book a hotel room.
- The user should receive a booking confirmation after completing the reservation.

3.Use Case Diagram:

The system includes the following actors:

- User

- Payment Gateway
- External Booking Providers

The system supports the following use cases:

- Search Hotels
- Apply Search Filters
- View Hotel Details
- Compare Hotel Prices
- Book a Hotel Room

The Use Case Diagram illustrates the interactions between actors and the system and shows the relationships between the use cases.
(The Use Case Diagram is attached in the document)

4. Functional Requirements:

FR-01: Search Hotels

- **Description:** Allows the user to search for hotels in a selected city.
- **Inputs:** City name, check-in date, check-out date.
- **Source:** User
- **Pre-condition:** User has internet access.
- **Post-condition:** A list of available hotels is displayed.
- **Output:** Hotel search results.

FR-02: Apply Search Filters

- **Description:** Allows the user to filter hotel search results based on specific criteria.
- **Inputs:** Price range, rating, number of stars.
- **Source:** User
- **Pre-condition:** Search results are available.
- **Post-condition:** Filtered search results are displayed.
- **Output:** Filtered list of hotels.

FR-03: View Hotel Details

- **Description:** Displays detailed information about a selected hotel.
- **Inputs:** Selected hotel.
- **Source:** User
- **Pre-condition:** Search results are available.
- **Post-condition:** Hotel details page is displayed.
- **Output:** Hotel information (images, services, ratings).

FR-04: Compare Hotel Prices

- **Description:** Compares hotel prices from multiple booking providers.
- **Inputs:** Selected hotel.
- **Source:** User / External Booking Providers
- **Pre-condition:** Hotel details are displayed.
- **Post-condition:** Price comparison results are shown.
- **Output:** List of hotel prices from different providers.

FR-05: Book a Hotel Room

- **Description:** Allows the user to book a hotel room and complete the payment process.
- **Inputs:** User details, payment information.
- **Source:** User / Payment Gateway
- **Pre-condition:** User is logged in and the selected room is available.
- **Post-condition:** Booking is successfully completed.
- **Output:** Booking confirmation message.

5. Sequence Diagram:

The Sequence Diagram illustrates the interaction sequence between the following entities:

- User
- Booking Page
- Booking Controller
- Rooms Database

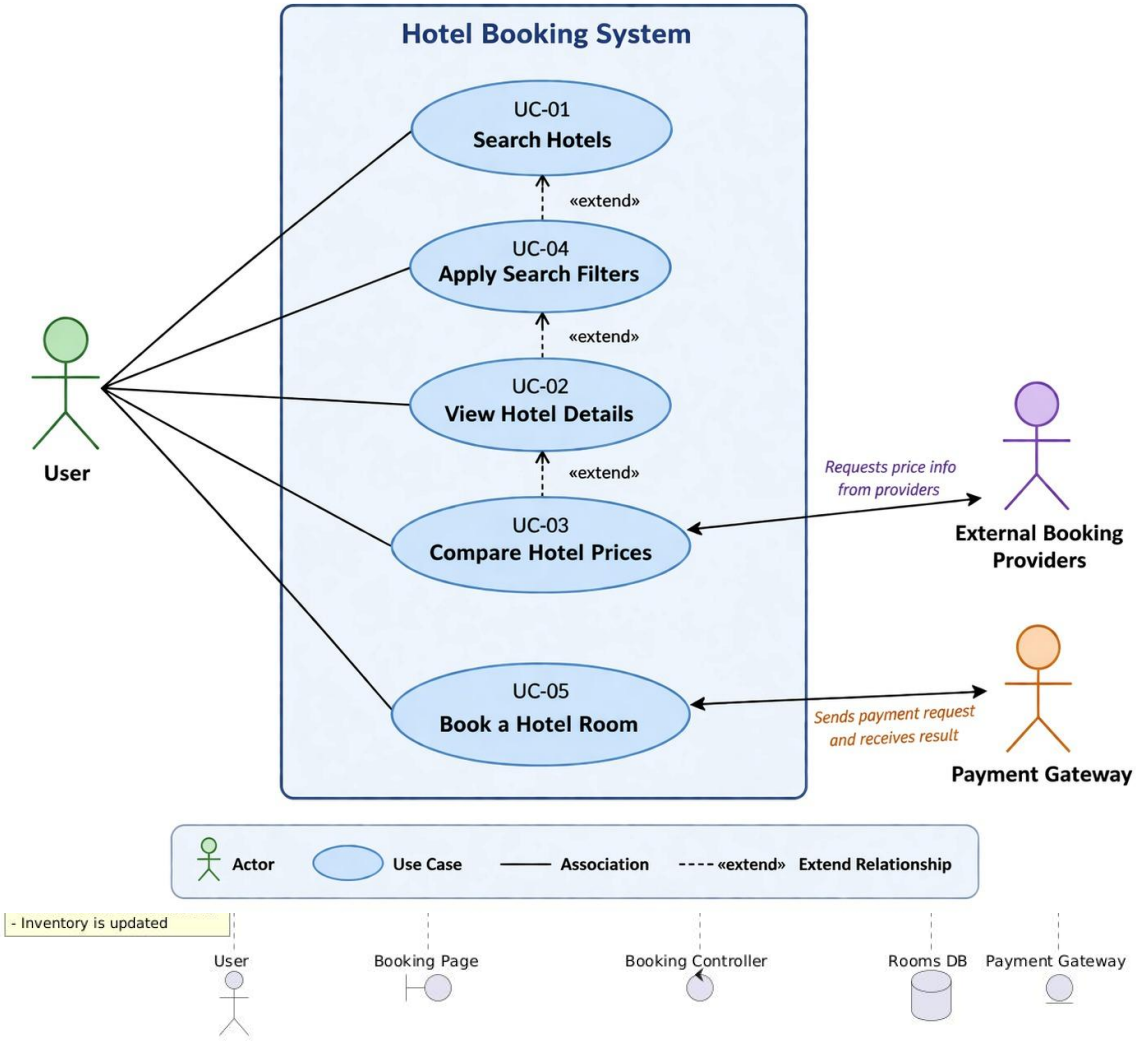
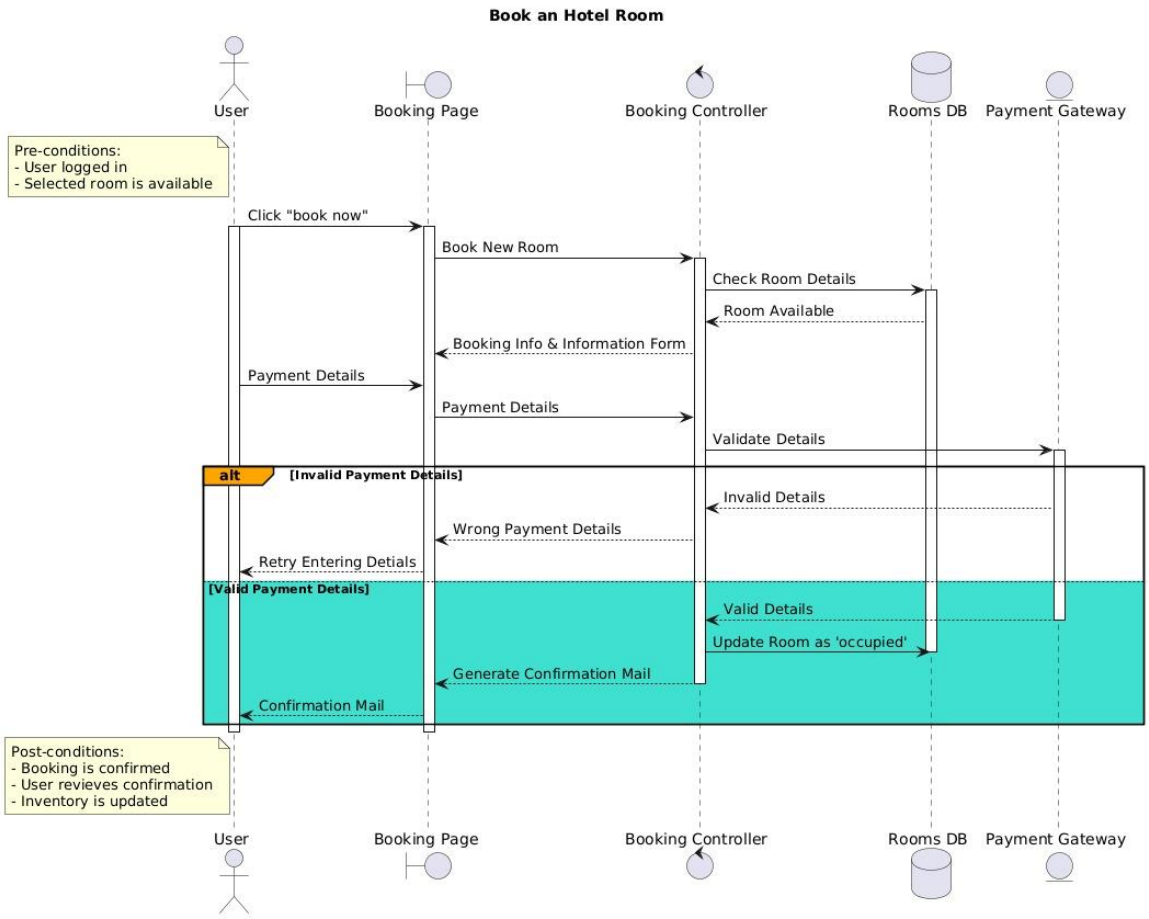
- Payment Gateway

It demonstrates the steps involved in booking a hotel room, starting from room selection and ending with payment confirmation and booking confirmation.
(The Sequence Diagram is included in the document)

6. Non-Functional Requirements

- **Performance:** The system should display search results within 3 seconds.
- **Availability:** The system should be available 99% of the time.
- **Security:** All payment data must be encrypted using secure communication protocols (HTTPS).
- **Usability:** The system should be easy to use and user-friendly.
- **Compatibility:** The system should work properly on major web browsers and devices

Use Case Diagram



UC 5 Sequence Diagram